

Comparison of Statistical Emulators for Parameter Estimation in Complex Systems

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Introduction

Inverse Problem Is Hard

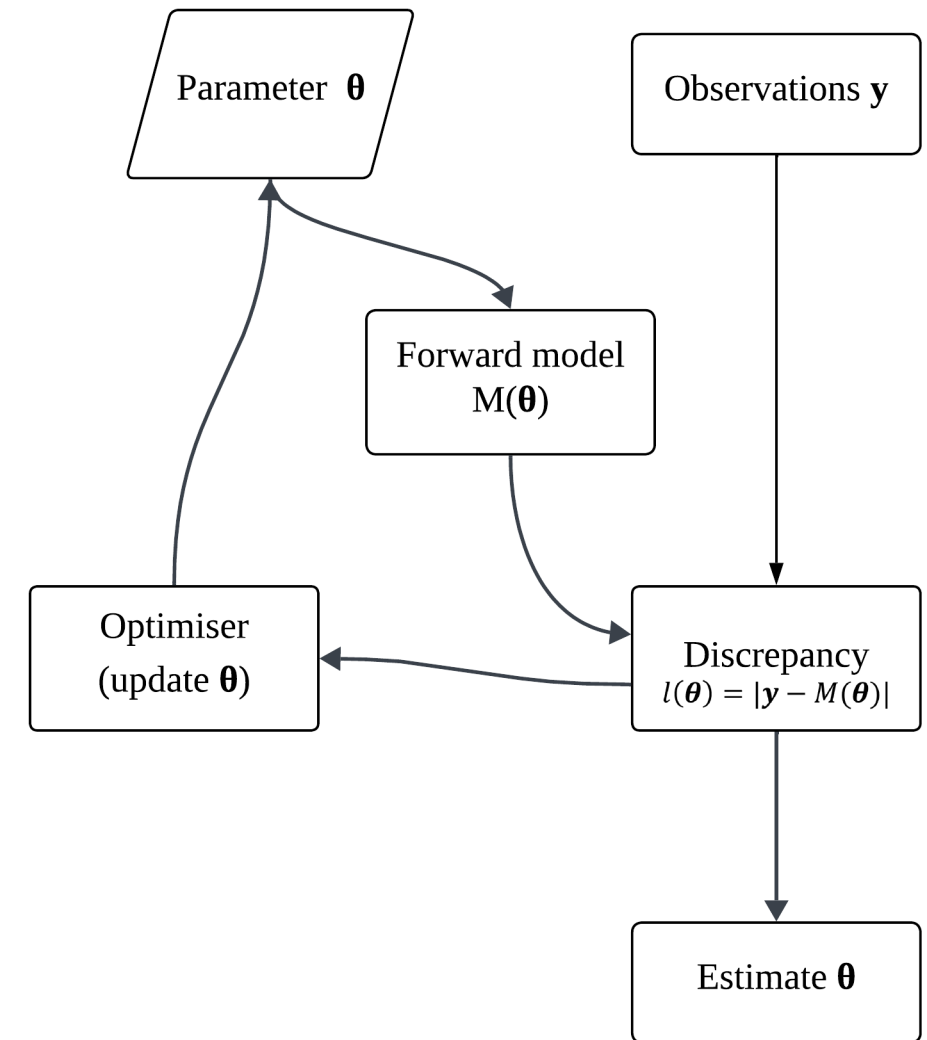
- Real-time decisions need fast parameter inference.
- Full simulators: minutes $\rightarrow$ hours per run.
- Hundreds–thousands of runs per inverse problem $\rightarrow$ **infeasible**.

Emulators as a Solution

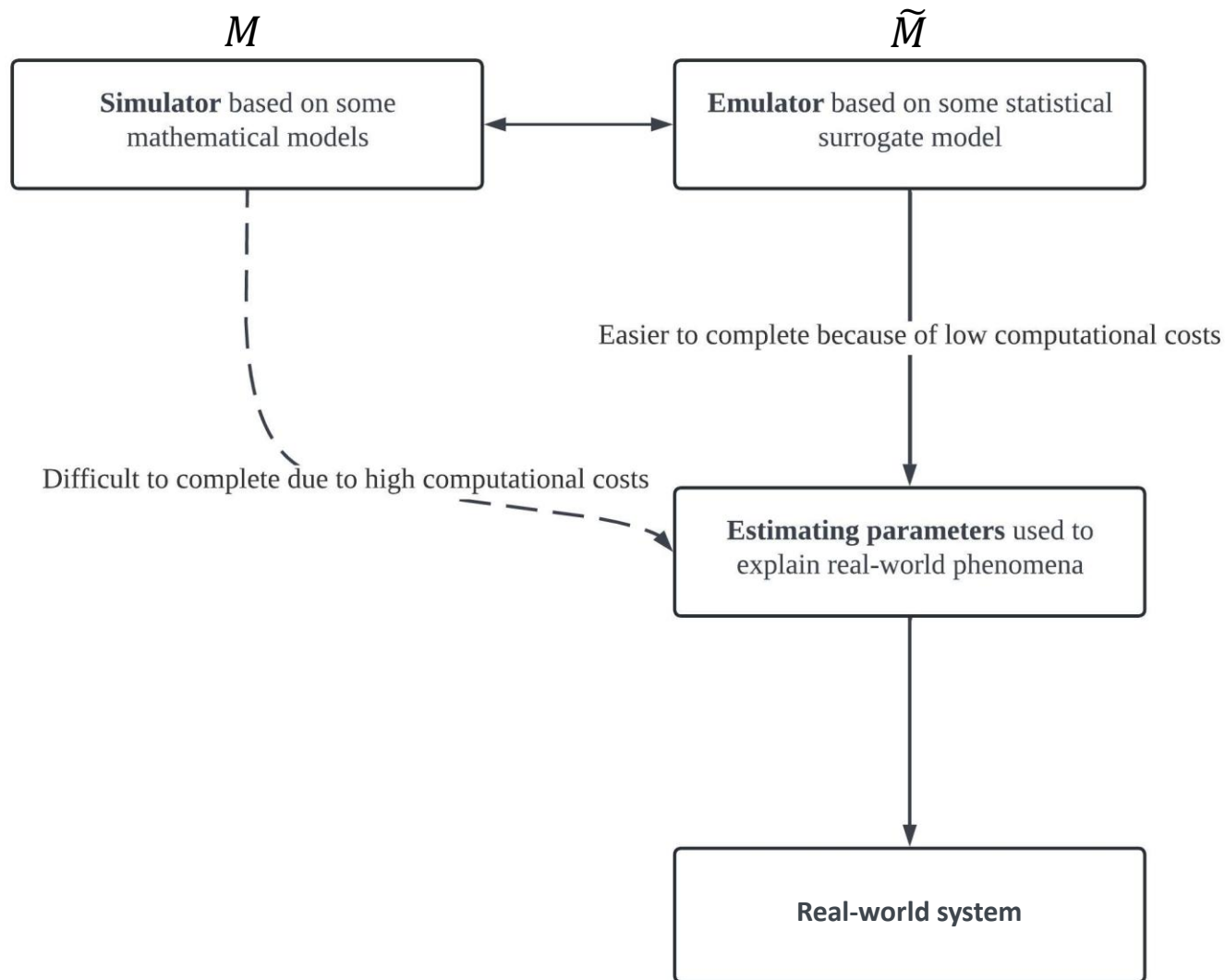
- **Statistical emulators** approximate the simulator.
- Train once on pre-computed data, then evaluate in seconds.

Key challenge:

- Which emulator architecture performs best under different data scenarios?



Workflow



Offline:

- Design the input parameter θ and generate data from simulator $M(\theta)$.
- Train emulator $\tilde{M}$.

Online:

- Use emulator $\tilde{M}$ to estimate $\hat{\theta} = \arg \min l(\theta | \tilde{M}, y)$.

Study Design

Inputs:

Sobol sequence to cover the parameter space.

Outputs:

Draw from multivariate normal distribution.

Factor varied	Levels tested	Fixed settings
Output correlation	Independent vs. Correlated	$d = 3, p = 6, n = 2024$
Output dimension	$p = 4,5,6,7,8$	$d = 5, n = 4048$
Training size	$n = 4048, 3036, 2024, 1024$	$d = 5, p = 6$

Emulators

Gaussian Processes:

- Standard Gaussian Processes (GP)
- Deep Kernel Learning Gaussian Processes (DKGP)
- Variational Gaussian Processes (VGP)
- Deep Gaussian Processes (DGP)

Neural Network :

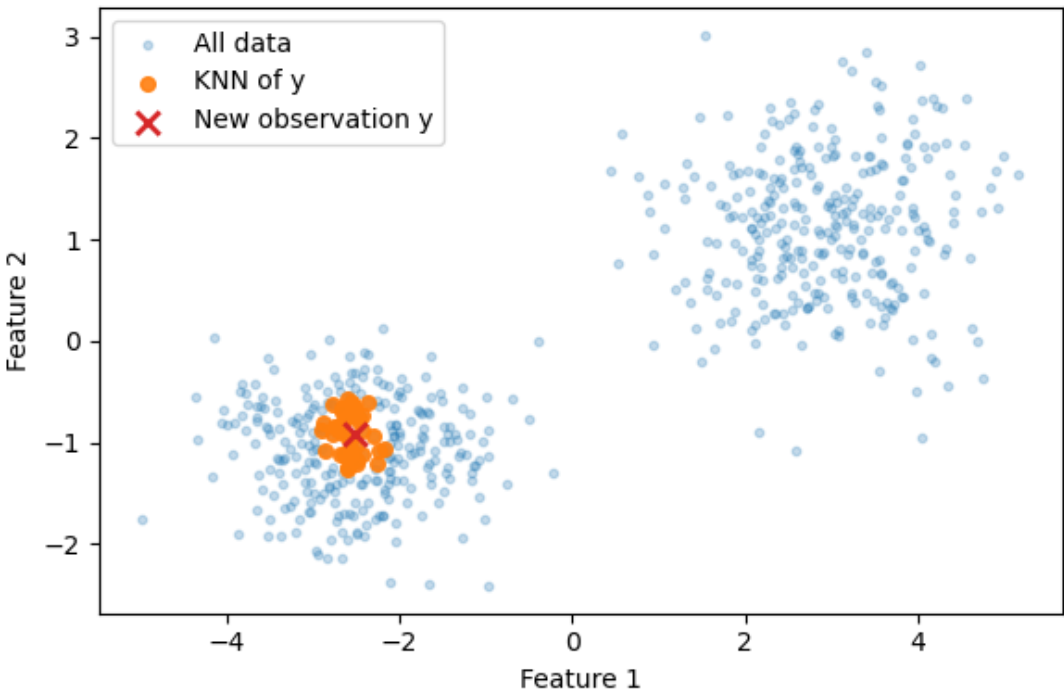
- Deep Neural Network (DNN)

Category	Single-output	Multi-output
Local	L.GP	L.MGP
	L.DKGP	L.DKMGP
Global	VGP	MVGP
		DGP
		DNN

Emulators

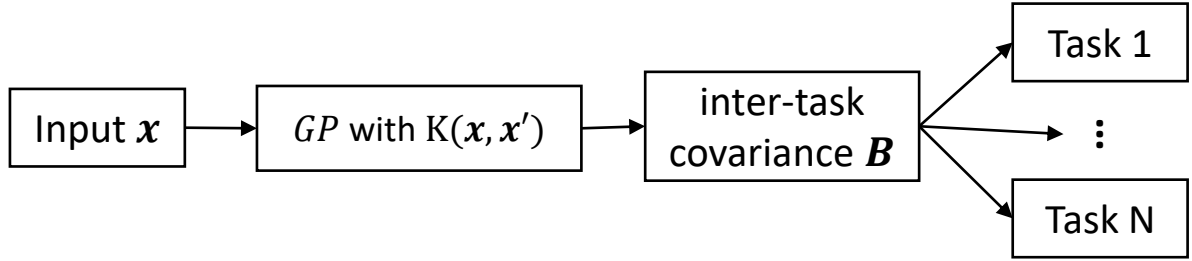
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Local & Global

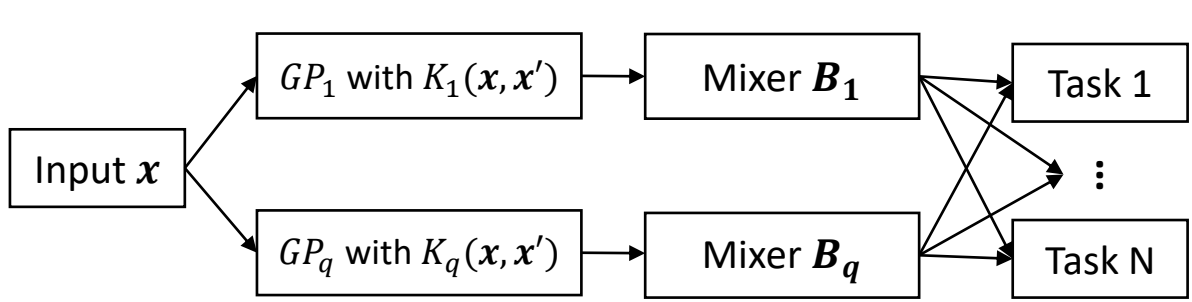


Single & Multi

➤ Intrinsic Coregionalisation Model (ICM)

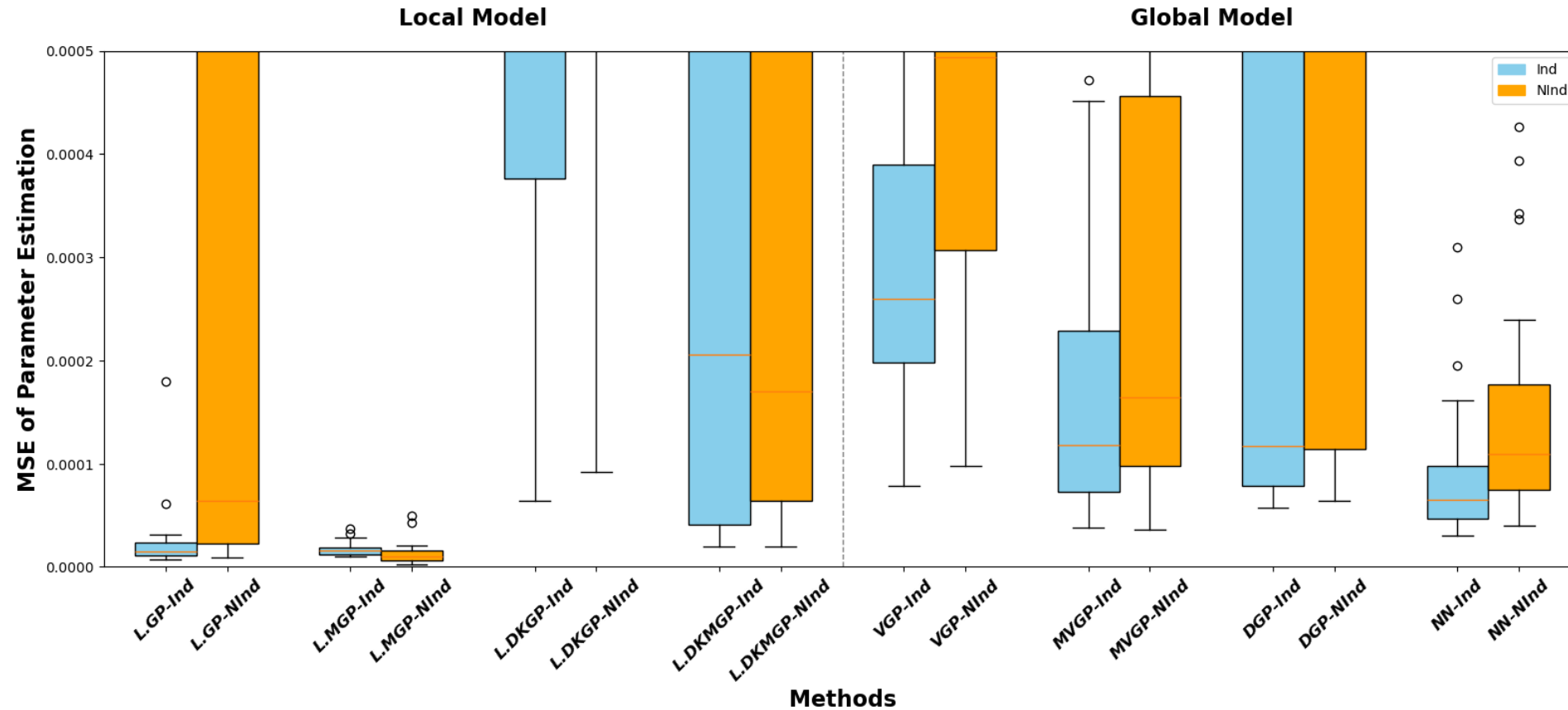


➤ Linear Model of Coregionalisation (LMC)



Results

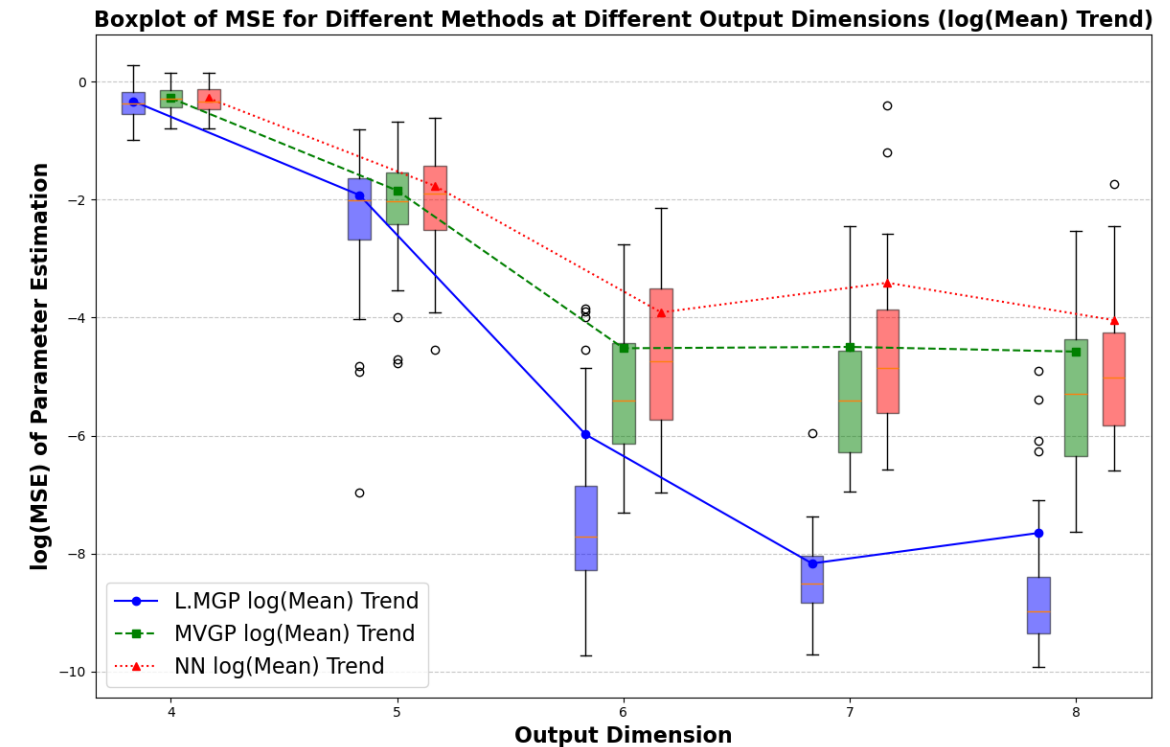
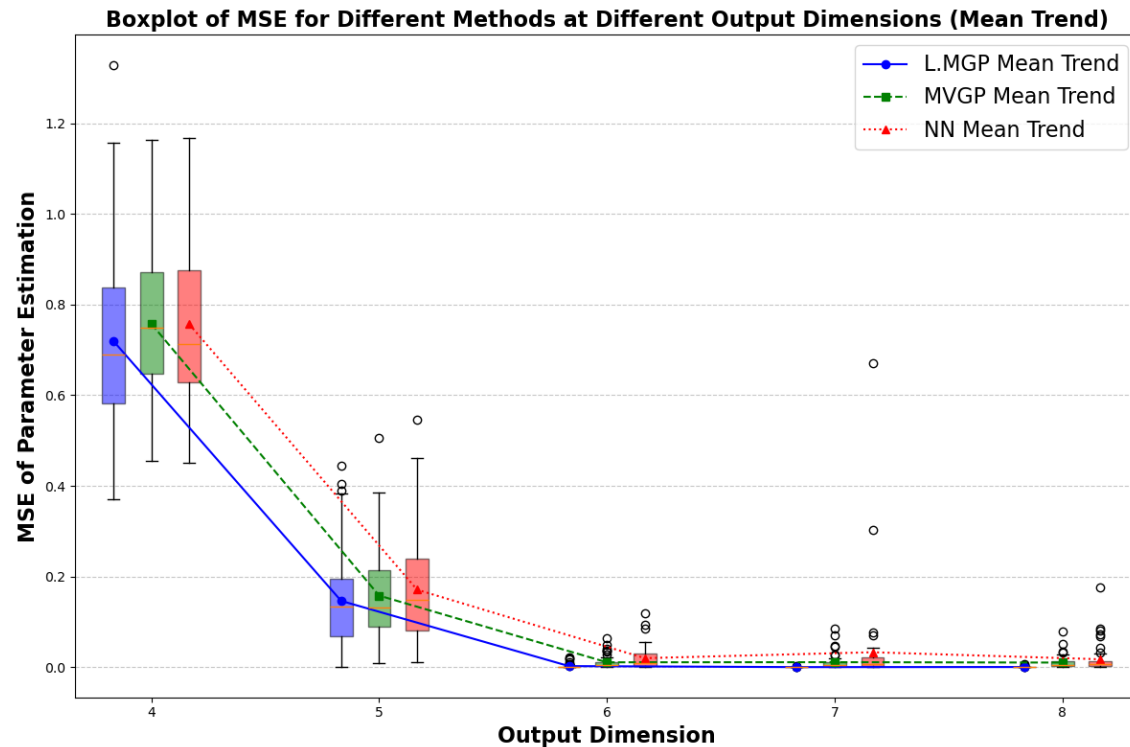
Experiment 1: Output Correlation



- Fix input dimension $d = 3$, outputs dimension $p = 6$.
- Multi-output methods are more accurate and stable with correlated outputs.
- Deep kernels are not consistently better than standard kernels in this setting.

Results

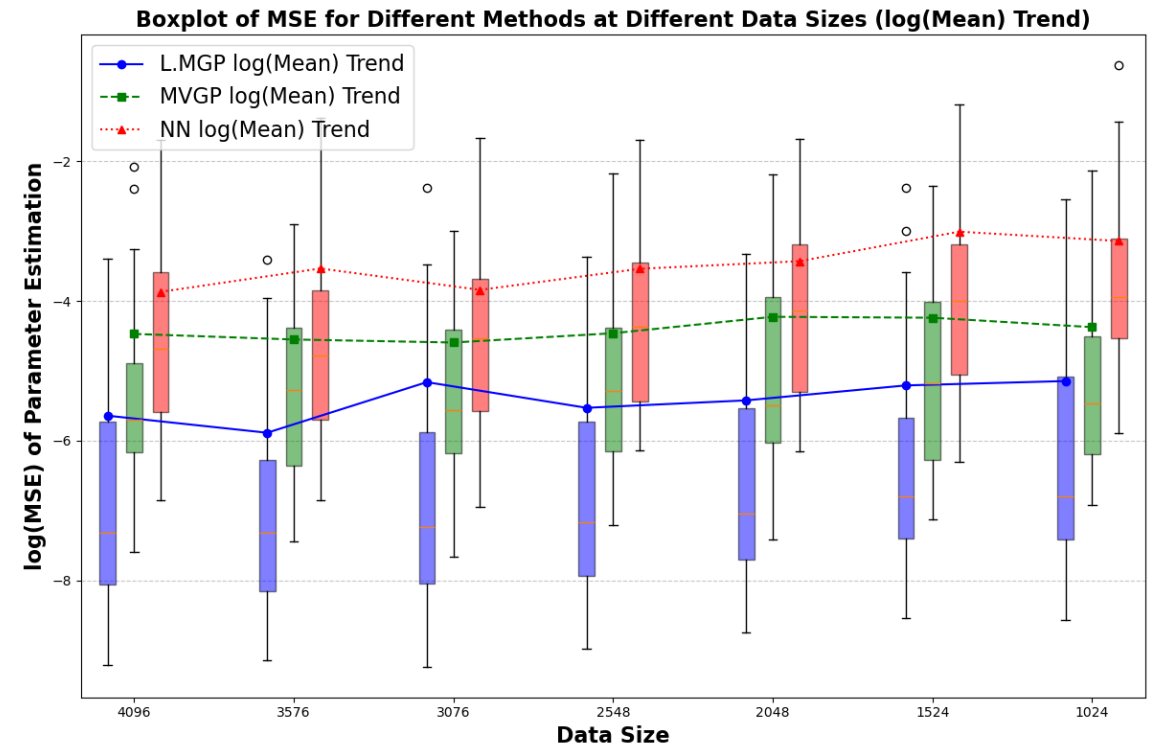
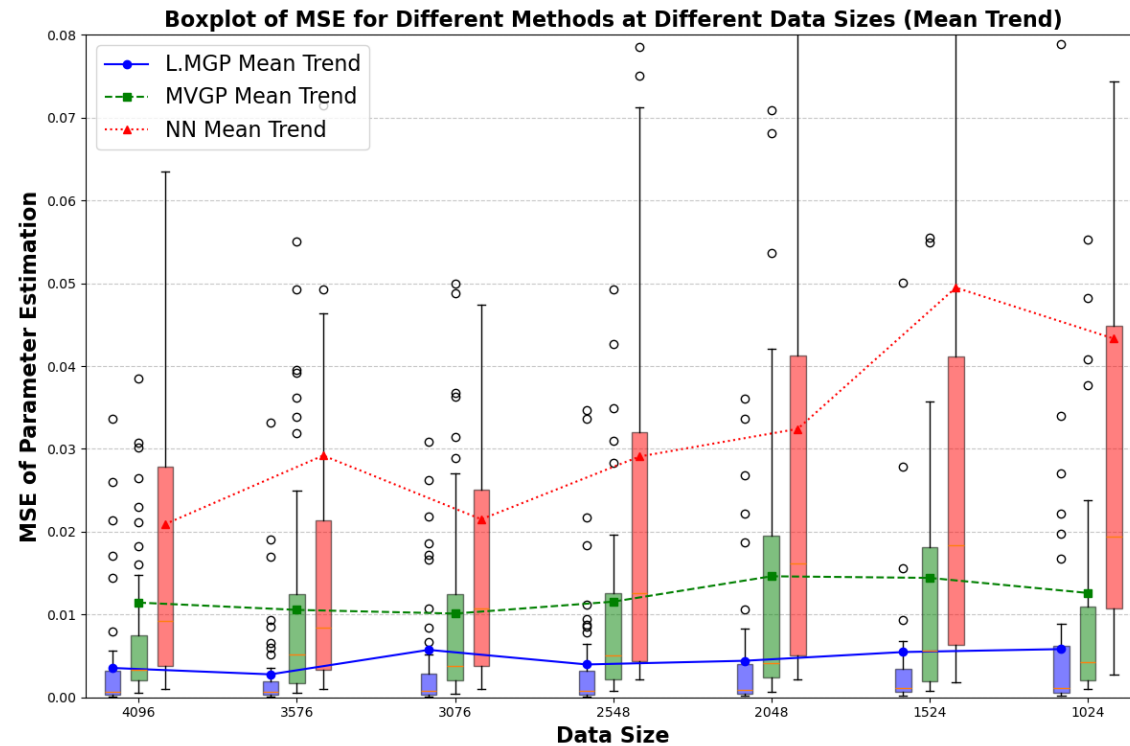
Experiment 2: Output Dimension



- Fix input dimension $d = 5$, increase outputs dimension p from $4 \rightarrow 8$
- Accuracy improves up to output $\approx$ input+1, then plateaus
- GP-based models outperform DNN once outputs $>$ inputs

Results

Experiment 3: Data Size



- Cut training data 4048 $\rightarrow$ 1024, fix input dimension $d = 5$, outputs dimension $p = 6$.
- GP models stay stable (small MSE shift)
- DNN variance increases $\rightarrow$ needs more data

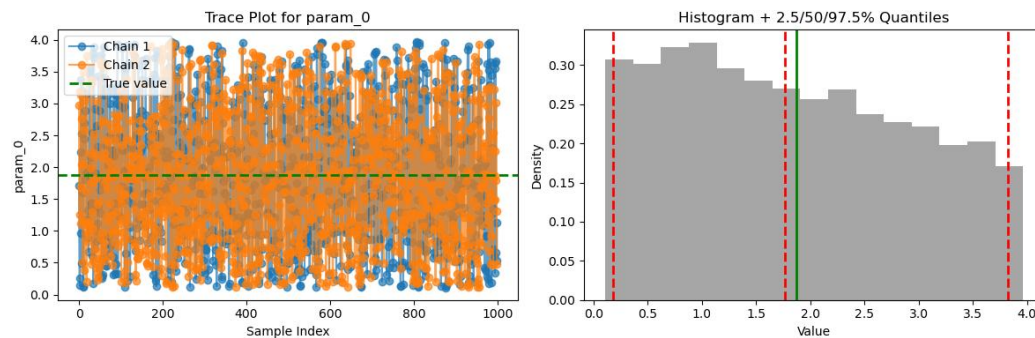
Conclusions & Next Steps

Conclusions

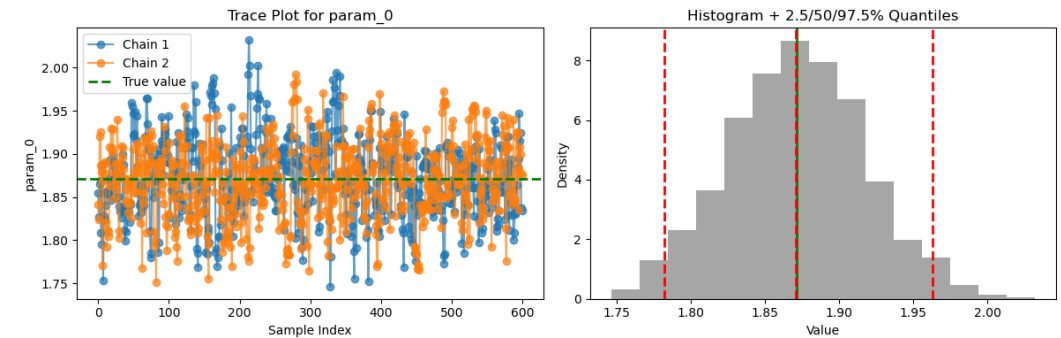
- If **global** and **local** achieve similar accuracy, prefer **global** for faster online use.
- **Multi-output** methods improve accuracy and stability in output correlation data size.

Future work

- Bayesian inference for uncertainty quantification
- Real cardiac models in clinic



L.MGP



MVGP

Thanks for listening!

